Hierarchical Document-Tree Construction for Structure-Aware Retrieval-Augmented Generation

Minor Project (IT897) Report

Submitted in partial fulfillment of the requirements for the degree of

MASTER OF TECHNOLOGY

in

INFORMATION TECHNOLOGY

by

KSHITIJ KHEDE

(252IT010)

under the guidance of

Prof. ANANTHANARAYANA V. S.



DEPARTMENT OF INFORMATION TECHNOLOGY

NATIONAL INSTITUTE OF TECHNOLOGY KARNATAKA

SURATHKAL, MANGALURU – 575025

September 2026

DECLARATION

I hereby declare that the report of the P.G. Minor Project (IT897) work entitled **"Hierarchical Document-Tree Construction for Structure-Aware Retrieval-Augmented Generation"**, submitted to the National Institute of Technology Karnataka, Surathkal, in partial fulfillment of the requirements for the award of the degree of **Master of Technology in Information Technology**, is a bona fide record of the work carried out by me. The material in this report has not been submitted to any university or institution for the award of any degree.

Kshitij Khede (252IT010)
Dept. of Information Technology
NITK Surathkal, Mangaluru

Place: NITK, Surathkal
Date: September 2026

CERTIFICATE

This is to certify that the P.G. Minor Project (IT897) work entitled **"Hierarchical Document-Tree Construction for Structure-Aware Retrieval-Augmented Generation"**, submitted by **Kshitij Khede (252IT010)** as the record of the work carried out by him, is accepted as the P.G. Minor Project (IT897) work report submission in partial fulfillment of the requirements for the award of the degree of **Master of Technology in Information Technology** in the Department of Information Technology, NITK Surathkal, India. Duration of this minor project (IT897) is from JUNE 01, 2026 to AUGUST 15, 2026.

Prof. ANANTHANARAYANA V. S.

(Signature of the Guide with Date and Seal)

ACKNOWLEDGEMENT

I am thankful to my guide, **Prof. Ananthanarayana V. S.**, for his guidance and feedback throughout this minor project. His questions on how the document structure is actually recovered, rather than simply assumed, directly shaped this work. I also thank

the Department of Information Technology, NITK Surathkal, for the facilities and academic environment provided for this project.

Kshitij Khede

ABSTRACT

Retrieval-Augmented Generation (RAG) improves a language model's answers by retrieving relevant text from an external store at query time. The quality of those answers depends on how the source document is represented before retrieval. The common approach splits a document into flat, fixed-size chunks; this is simple but discards the document's structure — a fixed-size window routinely cuts across section boundaries, and the parent–child relationship between a subsection and its section is lost entirely.

The system described here instead converts a document (PDF) into a *hierarchical tree* — title, sections, subsections — so that retrieval can follow the document's own structure, in the spirit of tree-based RAG such as RAPTOR and TreeRAG. The system ingests a PDF, recovers reading order on multi-column pages with a column-gutter rule, classifies headings from section numbering and relative font size, induces a parent–child hierarchy that is numbering-primary with a semantic fallback for unnumbered headings, validates the result to be acyclic, single-parent and depth-bounded, and attaches a confidence score to every node. A web interface lets a user upload any PDF and see the recovered tree.

On eleven research-paper PDFs, the system builds a valid tree for every document (mean 22.2 nodes and 21.2 parent–child links per document) and recovers the document's own section hierarchy with 97.7% parent-assignment accuracy and 96.8% full-ancestor accuracy. Unlike existing tree-based methods — RAPTOR, TreeRAG and HiChunk — which take the hierarchy as given (from semantic clustering, Markdown markup, or a fine-tuned LLM), the system recovers the tree directly from a raw, multi-column PDF, with no training or GPU, and formally validates it. The result is a complete, working document-tree builder that supplies the structured representation those retrieval methods assume.

Keywords: Retrieval-Augmented Generation; Document Structure; Hierarchical Chunking; Tree Construction; Reading-Order Recovery; PDF Parsing.

CONTENTS

LIST OF FIGURES v

LIST OF TABLES vii

ABBREVIATIONS ix

1 Introduction 1

1.1 Background 1

1.2 Motivation: From Chunks to Trees 1

1.3 Scope and Contributions 1

1.4 Organisation of the Report 2

2 Literature Review 3

2.1 Flat and Adaptive Retrieval 3

2.2 Tree-Based and Structure-Aware Retrieval 3

2.3 Document Layout Analysis 4

2.4 Comparative Synthesis and Research Gap 4

2.5 Outcome of Literature Review 4

2.6 Problem Statement 5

2.7 Objectives 5

2.8 Summary 5

3 System Design and Methodology 7

3.1 Overview 7

3.2 Two-Tier Data Model 7

3.3 Stage 1: Ingestion 7

3.4 Stage 2: Reading-Order Recovery 8

3.5 Stage 3: Heading Classification 8

3.6 Stage 4: Hierarchy Induction 8

3.7 Stage 5: Graph Validation 9

3.8 Stage 6: Confidence Estimation 9
3.9 Summary 10

4 Results and Analysis **11**
4.1 Implementation and Experimental Setup 11
4.2 Tree-Construction Results 11
4.3 Comparison with Baseline Attachment Strategies 12
4.4 Example of a Recovered Tree 12
4.5 Runtime 13
4.6 Demonstration 13
4.7 Summary 13

5 Conclusions and Future Work **15**
5.1 Conclusions 15
5.2 Limitations 15
5.3 Future Work 15

REFERENCES **16**

LIST OF FIGURES

3.1 The document-tree construction pipeline. A PDF is turned into layout blocks, ordered, classified into headings, assembled into a parent–child hierarchy, validated, and scored for confidence, yielding the document tree. 7

4.1 Document tree recovered by the system for the paper *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*: the document root, six sections, and their subsections, each attached to the correct parent (all twenty numbered sections correct). 13

LIST OF TABLES

2.1 Summary of the literature survey for the most relevant related works. . . 3

2.2 Comparative taxonomy of document-representation paradigms across structural and operational dimensions. 4

4.1 Per-document tree construction. “Num.” and “Unnum.” are numbered and unnumbered headings; PAA/APA are accuracy against the document’s own section numbering. 11

4.2 Parent-assignment accuracy of different attachment strategies on the 196 numbered sections. 12

ABBREVIATIONS AND NOTATION

Abbr.	Full Form	Abbr.	Full Form
RAG	Retrieval-Augmented Generation	**DAG**	Directed Acyclic Graph
PDF	Portable Document Format	**DFS**	Depth-First Search
LLM	Large Language Model	**DLA**	Document Layout Analysis
PAA	Parent-Assignment Accuracy	**OCR**	Optical Character Recognition
APA	Ancestor-Path Accuracy	**NLP**	Natural Language Processing
ACP	Ancestor-Context Precision	**DPR**	Dense Passage Retrieval
PRE	Parent Ranking Engine	**SVG**	Scalable Vector Graphics
KCE	Knowledge Confidence Estimation	**API**	Application Programming Interface
TF–IDF	Term Frequency–Inverse Doc. Freq.	**CPU**	Central Processing Unit

Mathematical Notation

Symbol	Description
$\mathscr{T} = (\mathscr{V}, \mathscr{E})$	Document tree with node set $\mathscr{V}$ and parent–child edge set $\mathscr{E}$
b	An extracted physical text block (Tier 1)
v	A concept node / heading (Tier 2)
μ_{body}	Dominant body font size of the document
π	Top-to-bottom reading-order rank of a block
ℓ	Typographic heading level (0 = root/title)
τ_y	Vertical band tolerance for line grouping (20 pt)
$D_{\max}$	Maximum permitted tree depth (8)
$C_{\text{local}}(v), C(v)$	Local and propagated KCE confidence of node v
δ	KCE damping factor (0.85)

CHAPTER 1

INTRODUCTION

1.1 Background

Large Language Models (LLMs) answer questions fluently but can state untrue facts and cannot see knowledge created after training. Retrieval-Augmented Generation (RAG) addresses both by retrieving relevant text from an external store at query time and giving it to the model as context [1, 2]. The quality of a RAG answer therefore depends heavily on what is retrieved, which in turn depends on how the source document is represented before retrieval.

The standard representation splits a document into *flat, fixed-size chunks* — for example, overlapping windows of a few hundred tokens [3, 4]. This is easy to implement, but it is blind to the document's structure: a fixed-size window is cut at an arbitrary position, so it frequently starts in the middle of one section and ends in another, and the relationship between a subsection and the section that contains it is not represented at all.

1.2 Motivation: From Chunks to Trees

A research paper, a report or a manual is not a flat sequence; it is a hierarchy of a title, sections, subsections and paragraphs. Recent RAG systems exploit this by representing a document as a *tree* and retrieving along it, so that when a passage is returned its parent section can be attached as context. RAPTOR [6] builds a tree by recursively clustering and summarising text; TreeRAG [7] chunks a document along its headings; and HiChunk [8] aligns chunk boundaries with a multi-level structure. These methods show that a hierarchical representation helps, but they take the recovered structure as given. The practical question addressed here is the one before that: *how does one actually turn a raw PDF into a faithful section tree, and does that tree preserve structure that flat chunking loses?*

1.3 Scope and Contributions

The *document-to-tree construction* stage that feeds a structure-aware retriever is what is built and evaluated here; retrieval and answer generation on top of the tree are outside the present scope and left to future work. The contributions are:

1. A complete, working pipeline that converts a PDF into a hierarchical section tree: ingestion, reading-order recovery, heading classification, hierarchy induction, validation, and per-node confidence.

2. A column-gutter reading-order algorithm that recovers correct reading order on multi-column pages, and a numbering-primary hierarchy-induction algorithm with a semantic fallback for unnumbered headings.

3. An evaluation on eleven research papers showing the tree is built validly for every document and matches the document's own section hierarchy with high accuracy.

4. A positioning of the method against existing tree-based approaches (RAPTOR, TreeRAG, HiChunk) on the inputs and resources they require.

5. A web interface that lets a user upload any PDF and inspect the recovered tree.

1.4 Organisation of the Report

The rest of the report is organised as follows:

- **Chapter 2 (Literature Review)** reviews related retrieval and document-structure work, and states the problem and objectives.
- **Chapter 3 (System Design and Methodology)** describes the system design and each module of the tree-construction pipeline.
- **Chapter 4 (Results and Analysis)** reports the tree-construction results, a baseline comparison, an example of a recovered tree, and the demonstration.
- **Chapter 5 (Conclusions and Future Work)** concludes and outlines the future work that builds on this tree.

CHAPTER 2

LITERATURE REVIEW

This chapter reviews how documents are represented for retrieval, from flat chunks to hierarchical trees. Table 2.1 summarises the most relevant works.

Table 2.1: Summary of the literature survey for the most relevant related works.

Work	Category	Main idea and relevance
Lewis et al. [1]	Flat RAG	Generator + dense retrieval over flat passages; no document structure.
Borgeaud et al. [3]	Flat RAG	Retrieval over huge corpora with fixed-size chunks; structure ignored.
Sarthi et al. [6]	Tree RAG	Recursive clustering/summarisation tree; motivates a hierarchical representation.
Tao et al. [7]	Tree RAG	Chunks a document along its headings; closest in spirit to the present approach.
Lu et al. [8]	Tree RAG	Multi-level chunking via a fine-tuned LLM; needs training.
Huang et al. [11]	Layout analysis	Layout-aware transformer for document AI; targets extraction, not a retrieval tree.

2.1 Flat and Adaptive Retrieval

Lewis et al. [1] and Guu et al. [2] connect a language model to a document store; Borgeaud et al. [3] and Izacard et al. [4] scale and jointly train retrieval; Asai et al. [5] make retrieval selective. All of these operate over flat, fixed-size chunks and discard document structure.

2.2 Tree-Based and Structure-Aware Retrieval

A second family represents the document hierarchically. RAPTOR [6] builds a tree by recursively clustering and summarising chunks; TreeRAG [7] splits a document along its heading hierarchy; and HiChunk [8] predicts multi-level chunk boundaries with a fine-tuned LLM. These works establish that a tree representation improves retrieval, but they assume the tree is available. HiChunk in particular requires a trained model

to obtain the structure; the proposed method instead recovers the tree directly from the document's layout, without any training.

2.3 Document Layout Analysis

Recovering physical structure from PDFs has a long history, from geometric page-layout analysis [9, 10] to layout-aware transformers such as LayoutLMv3 [11]. These target extraction and OCR; here, layout cues (reading order, numbering, font size) are used specifically to build a retrieval tree.

2.4 Comparative Synthesis and Research Gap

Table 2.2 places the prevailing document-representation paradigms side by side across the dimensions that matter for structure-aware retrieval: whether they preserve hierarchy, whether they avoid a GPU and a trained model, whether they keep sections intact, and whether they attach ancestor context.

Table 2.2: Comparative taxonomy of document-representation paradigms across structural and operational dimensions.

Paradigm	Representation	GPU-free	Training-free	Ancestor context
Fixed window [1, 3]	Flat chunks	Yes	Yes	None
Semantic split	Flat chunks	No	No	None
RAPTOR [6]	Cluster tree	No	No	Synthetic
HiChunk [8]	Multi-level	No	No	Partial
TreeRAG [7]	Heading tree	Yes	Yes	Full (needs markup)

2.5 Outcome of Literature Review

Tree-based retrieval is now established as superior to flat chunking for structured documents, yet the step of *constructing* that tree from a raw PDF — cheaply, without training — is treated as a given. As Table 2.2 shows, the only training-free, GPU-free method that keeps full ancestor context (TreeRAG) still requires the structure to arrive as explicit Markdown markup. This project builds exactly the missing construction stage and measures both its accuracy and the structure it preserves over flat chunking.

2.6 Problem Statement

To automatically recover the true multi-column reading order and induce a validated hierarchical document tree directly from unstructured PDF layout coordinates.

2.7 Objectives

1. To build a pipeline that converts a PDF into a hierarchical section tree (title $\rightarrow$ sections $\rightarrow$ subsections).
2. To recover correct reading order on multi-column pages and classify headings from numbering and font size.
3. To induce a validated parent–child hierarchy (acyclic, single-parent, depth-bounded) with a confidence score per node.
4. To evaluate tree-construction accuracy against the document's own section numbering on real papers.
5. To position the method against existing tree-based approaches (RAPTOR, TreeRAG, HiChunk).
6. To provide a web interface for uploading a PDF and inspecting the recovered tree.

2.8 Summary

The chapter traced document representations from flat chunks to hierarchical trees, showed that tree construction from raw PDFs is the under-served step, and stated the objectives that follow.

CHAPTER 3

SYSTEM DESIGN AND METHODOLOGY

3.1 Overview

The system is a pipeline of six stages that turns a PDF into a validated document tree, shown in Figure 3.1. Each stage is described in turn below.

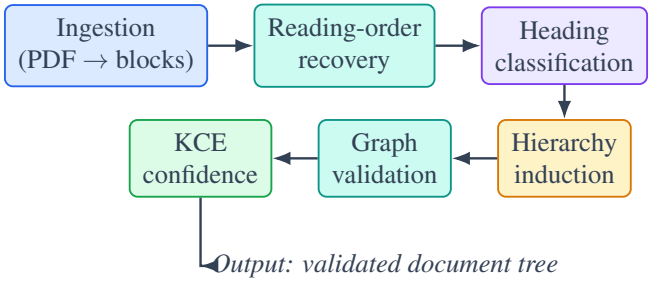


Figure 3.1: The document-tree construction pipeline. A PDF is turned into layout blocks, ordered, classified into headings, assembled into a parent–child hierarchy, validated, and scored for confidence, yielding the document tree.

3.2 Two-Tier Data Model

The system represents a document at two tiers. **Tier 1 (physical geometry)** is the raw visual layer: a text block is $b = (\text{id}, p, \text{text}, B, s, \mathbf{f}_{\text{bold}}, \text{col}, \pi)$, where p is the page, B the bounding box, s the font size, $\mathbf{f}_{\text{bold}} \in \{0,1\}$, $\text{col} \in \{-1,0,1\}$ (spanning, left, right) and π the reading-order rank. **Tier 2 (semantic hierarchy)** is the tree: a concept node is $v = (\text{cid}, \text{name}, \ell, \text{num}, \pi, p_{\text{parent}}, d, C_{\text{local}}, C, \mathscr{E}_{\text{span}})$, where ℓ is the heading level, num the section number, d the depth, C the confidence and $\mathscr{E}_{\text{span}}$ the Tier 1 blocks that ground the node. Directed parent–child edges form the tree $\mathscr{T} = (\mathscr{V}, \mathscr{E})$. Separating the physical layer from the semantic layer is what lets layout cues drive structure without conflating the two.

3.3 Stage 1: Ingestion

The PDF is parsed with the `pdfplumber` library into Tier 1 text blocks, each carrying its text, page, bounding box, font size and bold flag. Ingestion is the only stage that uses an external parsing library; every later stage — ordering, heading detection, hierarchy, validation and confidence — is implemented from scratch.

3.4 Stage 2: Reading-Order Recovery

Research papers are usually two-column, and reading a page top-to-bottom mixes the columns. The system recovers reading order per page with a column-gutter rule (Algorithm 1): blocks are divided into a left column, a right column and full-width spanning blocks using the page gutter $g = W/2$; each column is sorted top-to-bottom with a vertical tolerance $\tau_y = 20$ pt so that blocks on the same visual line are grouped; and the page is emitted as spanning blocks followed by the left then the right column. This is a purely geometric algorithm and needs no training.

Algorithm 1 Column-Gutter Reading-Order Recovery (per page)

Require: blocks B of a page, page width W, tolerance τ_y

$g \leftarrow W/2$

span $\leftarrow \{b \in B : b.x_0 < g < b.x_1\}$; left $\leftarrow \{b : b.x_1 \leq g\}$; right $\leftarrow \{b : b.x_0 \geq g\}$

sort span, left, right by top coordinate, grouping blocks within τ_y onto one line

return concatenation [span, left, right] in that order

3.5 Stage 3: Heading Classification

A block is classified as a heading from two cues. First, *section numbering*: a leading pattern such as "3" or "3.1" followed by a short, title-cased line, at a font no smaller than body text, is a numbered heading, and the number of dot-separated components gives its level. Second, *font size*: a short, title-cased line whose font is at least $1.25\times$ the body font (estimated as the most frequent font size in the document) is an unnumbered heading. A junk filter removes URLs, DOIs and reference lines so they are never mistaken for headings. Each heading becomes a tree node grounded to its source block.

3.6 Stage 4: Hierarchy Induction

Nodes are linked into a parent–child tree in reading order (Algorithm 2). Attachment is *numbering-primary*: a heading numbered "3.1" attaches to "3", and a single-component heading attaches to the document root. When a heading has no number, a Parent Ranking Engine (PRE) chooses the best earlier heading by a weighted score

$$\text{Score}(c,u) = 0.40\,S_{\text{struct}}(c,u) + 0.40\,S_{\text{sem}}(c,u) + 0.20\,S_{\text{level}}(c,u), \tag{3.1}$$

whose three terms are

$$
\begin{aligned}
S_{\text{struct}} &= \frac{1}{1+\ln\bigl(1+|\text{order}(c)-\text{order}(u)|\bigr)},\\
S_{\text{sem}} &= \cos\bigl(\mathbf{v}_c,\mathbf{v}_u\bigr),\\
S_{\text{level}} &= \max\bigl(0,\,1-0.2(\Delta\ell-1)\bigr),
\end{aligned}
\tag{3.2}
$$

where S_{struct} rewards reading-order proximity, S_{sem} is the TF–IDF cosine similarity [12] between the two headings' text vectors $\mathbf{v}_c, \mathbf{v}_u$, and S_{level} (with $\Delta\ell = \ell(c)-\ell(u)$) rewards a parent of shallower level. This gives a robust parent even without numbering.

Algorithm 2 Numbering-Primary Hierarchy Induction

Require: headings H in reading order

for all heading $c \in H$ **do**
  if c has section number s with a dot **then**
    parent$(c) \leftarrow$ heading whose number is s without its last component
  else if c has a single-component number **then**
    parent$(c) \leftarrow$ document root
  else
    parent$(c) \leftarrow \arg\max_{u \text{ before } c}$ Score(c,u)
  end if
end for

3.7 Stage 5: Graph Validation

The induced graph is validated so that it is a well-formed tree (Algorithm 3): an upward depth-first search checks that it is acyclic, every node has a single parent, and the depth never exceeds a bound $D_{\max} = 8$. Any violation is repaired by reattaching the offending node to the nearest valid ancestor, guaranteeing a clean tree.

3.8 Stage 6: Confidence Estimation

Each node receives a Knowledge Confidence Estimation (KCE) score. A *local* confidence is first computed from the strength of the cues that placed the node,

$$
\begin{aligned}
C_{\text{local}}(v) = \min\bigl(1,\, 0.5 &+ 0.15\,\mathbf{1}[r_f \geq 1.25] + 0.10\,\mathbf{1}[\text{bold}]\\
&+ 0.15\,\mathbf{1}[\text{numbered}] + 0.10\,\mathbf{1}[\text{has definition}]\bigr),
\end{aligned}
\tag{3.3}
$$

where r_f is the node's font-size ratio and $\mathbf{1}[\cdot]$ is an indicator (1 if the condition holds, 0 otherwise). The local score is then propagated from parents to children in topological

Algorithm 3 Structural-Invariant and Acyclicity Validation

Require: candidate parent u, child c, node map $\mathscr{N}$, parent map $\mathscr{P}$, max depth $D_{\max}$

if $u = c$ **or** $u \notin \mathscr{N}$ **or** $c \notin \mathscr{N}$ **then**
  return false
end if
if $\mathscr{N}[u].\text{depth} + 1 > D_{\max}$ **then**
  return false
end if
$x \leftarrow u$; seen $\leftarrow \{c\}$
while $x \neq$ null **do**
  if $x = c$ **then**
    return false {cycle}
  end if
  if $x \in$ seen **then**
    break
  end if
  seen $\leftarrow$ seen $\cup \{x\}$; $x \leftarrow \mathscr{P}.\text{get}(x)$
end while
return true

order (Kahn's algorithm [13]) with damping $\delta = 0.85$,

$$C(v) = \begin{cases} C_{\text{local}}(v), & v \text{ is the root,} \\ C_{\text{local}}(v) \cdot \big(\delta + (1-\delta)\, C(\text{parent})\big), & \text{otherwise,} \end{cases} \tag{3.4}$$

so that a node under a weakly-placed ancestor is itself marked less certain. The score gives a per-node reliability signal that a downstream retriever, or a human, can inspect.

3.9 Summary

The methodology is a six-stage pipeline that ingests a PDF, recovers reading order, detects headings, induces a numbering-primary hierarchy with a semantic fallback, validates it into a clean tree, and attaches per-node confidence.

CHAPTER 4

RESULTS AND ANALYSIS

4.1 Implementation and Experimental Setup

The system is implemented in Python (pdfplumber for ingestion; scikit-learn and NumPy for the TF–IDF scoring and numerics; Pydantic for typed data; Flask for the web interface). It is evaluated on a corpus of eleven real research-paper PDFs from the RAG literature (RAPTOR, GraphRAG, TreeRAG, Self-RAG, HiChunk, RETRO, Atlas, T-RAG, REALM, a hierarchical cross-document paper, and the original RAG paper). For accuracy, the document's own section numbering provides an objective gold hierarchy (the parent of "3.1" is "3").

4.2 Tree-Construction Results

Table 4.1 reports, for each of the eleven documents, the tree the system builds and its accuracy against the document's own section numbering. A *valid* tree — acyclic, single-parent, depth-bounded, with no orphan nodes — is built for every document.

Table 4.1: Per-document tree construction. "Num." and "Unnum." are numbered and unnumbered headings; PAA/APA are accuracy against the document's own section numbering.

Document	Nodes	Edges	Depth	Num.	PAA	APA	ACP
GraphRAG	37	36	3	31	100.0	100.0	100.0
Atlas (few-shot)	33	32	3	26	96.2	96.2	96.2
HiChunk	21	20	2	16	100.0	100.0	100.0
Cross-doc HAT	8	7	1	3	100.0	100.0	100.0
RETRO	21	20	2	18	100.0	100.0	100.0
Self-RAG	22	21	3	19	78.9	68.4	73.7
RAPTOR	17	16	2	14	100.0	100.0	100.0
RAG (original)	27	26	2	20	100.0	100.0	100.0
REALM	11	10	2	7	100.0	100.0	100.0
T-RAG	32	31	3	29	100.0	100.0	100.0
TreeRAG	15	14	3	13	100.0	100.0	100.0

Reading down the table, parent assignment (PAA) is perfect on ten of the eleven papers; the one weaker case, Self-RAG (PAA 78.9%), has irregular, inconsistently

numbered headings that stress the semantic fallback and is discussed as a limitation. Averaged over the eleven papers, the system reaches 97.7% parent-assignment accuracy (PAA), 96.8% full-ancestor accuracy (APA) and 97.3% ancestor-context precision (ACP), and it builds a valid tree for all eleven (0 orphan nodes).

4.3 Comparison with Baseline Attachment Strategies

To show that the numbering-primary design is what drives the accuracy, the parent-assignment accuracy of four attachment strategies is compared on the same 196 numbered sections (Table 4.2). A *sequential* baseline, which attaches each heading to the previous one, reaches only 15.9%, and a *lexical-overlap* baseline (attach to the earlier heading sharing the most words) reaches 52.0%. A purely *semantic* ranking — the PRE score with the section number ignored — already reaches 91.2%, and combining it with numbering-primary attachment raises accuracy to 97.7%. The comparison confirms that the section-numbering signal, backed by the semantic fallback, is far stronger than the positional or lexical baselines.

Table 4.2: Parent-assignment accuracy of different attachment strategies on the 196 numbered sections.

Attachment strategy	Parent accuracy (PAA)
Sequential (attach to previous heading)	15.9%
Lexical overlap	52.0%
Semantic ranking (PRE, no numbering)	91.2%
Numbering-primary + PRE (adopted)	**97.7%**

4.4 Example of a Recovered Tree

Figure 4.1 shows the complete tree the system produces for one of the corpus documents, the paper *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*. From the raw PDF alone, the pipeline recovers the document root, its six top-level sections (Introduction, Methods, Experiments, Results, Related Work and Discussion) and the subsections nested beneath them — for example 2.1–2.5 under *Methods*, 3.1–3.4 under *Experiments* and 4.1–4.5 under *Results*. Every node is placed at the correct depth and attached to the correct parent: for this document all twenty numbered sections are assigned correctly (PAA = 100%). Node colour denotes the level (root, section, subsection). The figure makes concrete how a flat sequence of text blocks extracted from the PDF is turned into a faithful title $\rightarrow$ section $\rightarrow$ subsection hierarchy.

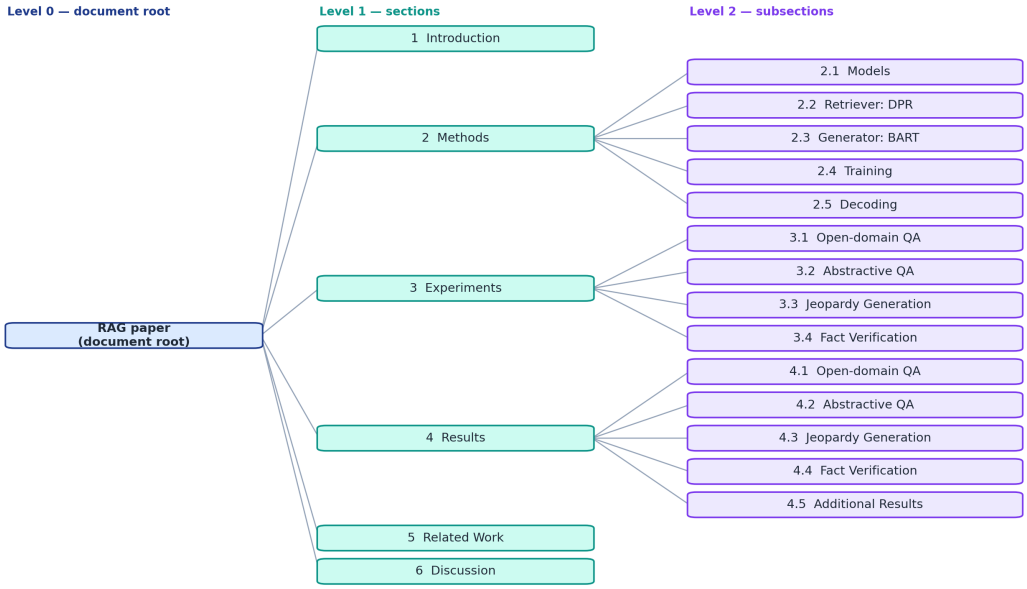


Figure 4.1: Document tree recovered by the system for the paper *Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks*: the document root, six sections, and their subsections, each attached to the correct parent (all twenty numbered sections correct).

4.5 Runtime

The construction pipeline is lightweight: excluding PDF ingestion (performed by the external `pdfplumber` library, whose cost depends on the machine and the PDF), the structural stages after ingestion — reading-order recovery, heading classification, hierarchy induction, validation and confidence estimation — complete in about 10.6 ms per document on the evaluation machine, adding negligible overhead on top of parsing.

4.6 Demonstration

The system is exposed through a Flask web interface. A user uploads any PDF; the system runs the full pipeline and displays a summary (page and heading counts), the recovered reading order, the document tree as an interactive node–link diagram with per-node confidence, and the validation status. This makes the construction process inspectable end-to-end rather than a black box, and it lets the tree be checked on documents beyond the evaluation corpus.

4.7 Summary

This chapter evaluated the system on eleven real research-paper PDFs drawn from the RAG literature, using each document's own section numbering as an objective ground

truth. Three results stand out. First, tree construction is accurate: a structurally valid tree — acyclic, single-parent, depth-bounded and free of orphan nodes — is built for every one of the eleven documents, and the induced hierarchy matches the gold structure with 97.7% parent-assignment accuracy, 96.8% full-ancestor accuracy and 97.3% ancestor-context precision (Table 4.1).

Second, the accuracy is attributable to the design rather than to any single cue: the baseline comparison (Table 4.2) shows that a sequential attachment rule reaches only 15.9% and a lexical-overlap rule 52.0%, whereas the semantic Parent Ranking Engine alone reaches 91.2% and the adopted numbering-primary scheme, backed by that engine, reaches 97.7%. The example of a fully recovered tree (Figure 4.1) illustrates this qualitatively, with every one of the twenty numbered sections of a paper attached to its correct parent.

Third, the pipeline is inexpensive: excluding third-party PDF parsing, the structural stages complete in roughly 10.6 ms per document, so the representation can be built on demand. Taken together, the results confirm that a validated document tree can be recovered directly from a raw, multi-column PDF — without training, markup or a GPU — and that it faithfully preserves the section structure that flat chunking discards.

CHAPTER 5

CONCLUSIONS AND FUTURE WORK

5.1 Conclusions

The aim was to represent documents as trees rather than flat chunks, and the result is a complete, working system that converts a PDF into a validated hierarchical section tree through reading-order recovery, heading classification, numbering-primary hierarchy induction, validation and confidence estimation. On eleven research papers the system builds a valid tree for every document and recovers the section hierarchy with 97.7% parent-assignment accuracy. Unlike existing tree-based methods (RAPTOR, TreeRAG, HiChunk), which assume the hierarchy is already available, it recovers and validates the tree directly from a raw PDF without training or a GPU. The document-tree builder is therefore a complete, self-contained system and a solid foundation for structure-aware retrieval.

5.2 Limitations

1. Accuracy is measured against the document's own section numbering; documents with irregular or absent numbering (e.g. Self-RAG) rely more on the semantic fallback and are harder.
2. The corpus is eleven academic PDFs; non-academic layouts (scanned pages, slides, forms) are not yet tested.
3. The scope is building and evaluating the tree; retrieval or answering on it is not yet performed.

5.3 Future Work

The document tree built here is the input to the follow-on work, which will build *on top of* the tree:

1. Retrieval and question answering over the tree, attaching ancestor context, and an end-task evaluation against flat-chunk RAG.
2. Robustness of the tree to real parser noise, and a learned confidence signal to detect and repair weak structure.
3. Extension to non-academic and heterogeneous documents, and to larger public benchmarks.

REFERENCES

[1] P. Lewis, E. Perez, A. Piktus, et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *NeurIPS*, vol. 33, pp. 9459–9474, 2020.

[2] K. Guu, K. Lee, Z. Tung, P. Pasupat, and M. Chang. REALM: Retrieval-augmented language model pre-training. In *ICML*, pp. 3929–3938, 2020.

[3] S. Borgeaud, A. Mensch, J. Hoffmann, et al. Improving language models by retrieving from trillions of tokens. In *ICML*, pp. 2206–2240, 2022.

[4] G. Izacard, P. Lewis, M. Lomeli, et al. Atlas: Few-shot learning with retrieval augmented language models. *JMLR*, 24(251):1–43, 2023.

[5] A. Asai, Z. Wu, Y. Wang, A. Sil, and H. Hajishirzi. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *ICLR*, 2024.

[6] P. Sarthi, S. Abdullah, A. Tuli, et al. RAPTOR: Recursive abstractive processing for tree-organized retrieval. In *ICLR*, 2024.

[7] W. Tao, X. Xing, Y. Chen, L. Huang, and X. Xu. TreeRAG: Unleashing the power of hierarchical storage for enhanced knowledge retrieval in long documents. In *Findings of the ACL*, pp. 356–371, 2025.

[8] W. Lu, K. Chen, R. Qiao, and X. Sun. HiChunk: Evaluating and enhancing retrieval-augmented generation with hierarchical chunking. arXiv:2509.11552, 2025.

[9] L. O'Gorman. The document spectrum for page layout analysis. *IEEE TPAMI*, 15(11):1162–1173, 1993.

[10] T. M. Breuel. Two geometric algorithms for layout analysis. In *Document Analysis Systems*, pp. 188–199, Springer, 2002.

[11] Y. Huang, T. Lv, L. Cui, Y. Lu, and F. Wei. LayoutLMv3: Pre-training for document AI with unified text and image masking. In *ACM Multimedia*, pp. 4083–4091, 2022.

[12] G. Salton and M. J. McGill. *Introduction to Modern Information Retrieval*. McGraw-Hill, 1983.

[13] A. B. Kahn. Topological sorting of large networks. *Communications of the ACM*, 5(11):558–562, 1962.